

What is a V-tail?

You must have heard of tricopters, quadcopters, hexacopters and octocopters that have three, four, six and eight propellers, respectively. V-tail is a drone that lies between a tricopter and a quadcopter, in that it has four motors but the rear propellers are turned in the shape of a V to provide for a very acrobatic flight.

Trying to make some sense

Without proper input to the drone (or its pilot), your drone is not going to reliably fly anywhere. There is quite a wide range of sensors that you can choose from to allow your drone to make sense of its surroundings.

Can I see. Drones also stand to gain from technologies like Intel's depth-sensing camera technology from 2014 called RealSense 3D, which has the potential to enable holographic navigation and real-time environment monitoring.

Sony's Exmor R CMOS sensors are used in drones as these excel at recording in low-light conditions and are also able to record smoother action sequences while preventing smear from strong light sources such as streetlights.

DJI, the current king of consumer drones, uses Lightbridge technology that enables streaming video at 760p without latency to the pilot, while the drone itself records video at 1080p full-high-definition.

Drones like Inspire 1 can also shoot 4K videos and feature rotors that can be lifted out of the view of the drone's camera, thus providing a 360-degree view. It is no surprise that this is the drone that lives on-board Etos car we mentioned earlier.

Considering the rise in solar energy installations in India, drones designed to detect small defects on photovoltaic modules are a very good application. Existing drones can also be fitted with payloads for this application, by pairing an infrared (IR) camera like FLIR TAU 640

About batteries and drones

Batteries used in unmanned aerial vehicles (UAVs) are now almost exclusively lithium-polymer. Lead-acid is simply not an option and nickel-metal hydride/nickel-cadmium batteries are still too heavy for their capacity and often cannot provide the high discharge rates needed in drones. Lithium-polymer offers high capacity with low weight, and high discharge rates. Its downsides are comparatively higher costs and continued safety issues. Lithium-ion batteries have higher discharge rates than lithium-polymer ones, but these cannot provide the high discharge rates needed in UAVs.

Lithium-polymer batteries used in drones are rated for 20C or more, where C is the capacity of the battery pack. The battery's voltage should correspond with the motors we choose in our drone. Typically, a 3-cell lithium-polymer pack is used, whose nominal voltage is $(3 \times 3.7) = 11.1V$, with capacity depending on the size of the UAV.

Small battery packs can be of 0.1Ah (100mAh), though battery packs for medium-size drones are in the range 2Ah-3Ah (2000mAh-3000mAh). A 10C discharge rating for a drone battery pack of 2Ah means that the pack can be safely discharged at $10 \times 2 = 20A$ current. The higher the capacity, the longer the flight time, or the heavier the pack can be.

Average flight time for a UAV is 15 to 20 minutes since it is always fighting against gravity. A challenge while charging a lithium-polymer battery pack is that it should not be overcharged nor should it be discharged below 3V per cell. Proper chargers for lithium-polymer ensure all cells in series are balanced else the pack can become unstable and dangerous. Depending on how these are used, most lithium-polymer batteries typically do not last longer than 300 charge cycles. Leaving these around on a full or depleted charge all the time, running these completely dead or exposing these to high temperatures will shorten their lifespan dramatically.

—Vikas Kumar Thawani, analog applications engineer, Texas Instruments India, and Naaveen Bevara, analog applications engineer, Texas Instruments India

Twitter: Is that a social network or a drone?

Twitter, the popular social network located at www.twitter.com, has been granted a patent for a drone by United States Patent & Trademark Office. The patent allows for a drone to be controlled via a messaging app that democratises control over the drone to a larger group of people (probably using Twitter's service).

and AscTec IR RAW data logger that can be used with German AscTec Falcon 8 drone. Falcon 8, considering its use in critical applications, comes with flight-control systems and electronics that deliver three times redundancy.

Where am I. Vision positioning is another technology used by DJI drones that lets the drone have high hover accuracy by using ultrasonic sensors. This lets a drone hold its position without using a global positioning system (GPS). DJI's Phantom drone also comes with FailSafe, which allows it to return back home in case it loses connection with the controller.

Intelligent orientation control allows easier control of the drone for the pilot as it considers forward direction of the drone to be forward direction of the pilot and not the

direction in which the drone's nose is pointing.

Newer drones have also taken the next step forward for positioning and have started to use Russian GLObal Navigation Satellite System (GLONASS) in addition to American GPS. Integration of better navigation systems into newer drones has also allowed these to be set to fly to a particular location, capture footage and then return back home autonomously.

Am I intelligent. Talking about autonomous drones, Zano is the result of a very successful Kickstarter project that raised almost ₹ 230 million. Small enough to fit in the palm of your hand, Zano drone comes with a smart piloting system that lets beginners easily pilot the drone. It also features IR obstacle-avoidance as well as echo-sounding